



Complex Numbers · Worksheet 1

Core Pure 2 Chapter 1: Complex Numbers. Answer all questions. Show your working clearly. Total: 87 marks. Time: 110 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are four sections, one for each topic in Chapter 1. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Give exact answers where possible. Arguments are in radians, in the interval $-\pi < \theta \leq \pi$ unless stated otherwise.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A De Moivre's theorem and powers	19	
B Using de Moivre for trigonometric identities	22	
C n th roots of a complex number	22	
D Roots of unity and their geometry	19	
Roar-level challenge	5	
Total	87	

Section A De Moivre's theorem and powers

19 marks

1. Use de Moivre's theorem to express $\left(\cos \frac{\pi}{9} + i \sin \frac{\pi}{9}\right)^6$ in the form $a + bi$, where a and b are exact real numbers. [2]

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2. Simplify $\frac{(\cos 3\theta + i \sin 3\theta)^4}{(\cos \theta - i \sin \theta)^2}$, giving your answer in the form $\cos n\theta + i \sin n\theta$. [2]

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3. By first writing $1 + \sqrt{3}i$ in modulus–argument form, use de Moivre's theorem to find $(1 + \sqrt{3}i)^9$. [3]

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4. Find, in the form $a + bi$, the exact value of [4]

$$\frac{(\sqrt{3} + i)^7}{(1 - i)^4}.$$

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5. (a) Prove by induction that, for all $n \in \mathbb{Z}^+$,

[5]

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta.$$

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(b) Hence find the smallest positive integer n for which $(\sqrt{3} + i)^n$ is real and positive, and find the value of $(\sqrt{3} + i)^n$ for this n .

[3]

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(Total for Question 5 is 8 marks)

**Section B Using de Moivre for trigonometric identities**

22 marks

6. Given that $z = \cos \theta + i \sin \theta$, show that $z^n + \frac{1}{z^n} = 2 \cos n\theta$. [2]

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7. Use de Moivre's theorem to show that $\cos 3\theta \equiv 4 \cos^3 \theta - 3 \cos \theta$. [3]

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8. By considering $\left(z + \frac{1}{z}\right)^4$, where $z = \cos \theta + i \sin \theta$, show that [3]

$$\cos^4 \theta \equiv \frac{1}{8} \cos 4\theta + \frac{1}{2} \cos 2\theta + \frac{3}{8}.$$

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9. Use de Moivre's theorem to show that [4]

$$\tan 3\theta \equiv \frac{3 \tan \theta - \tan^3 \theta}{1 - 3 \tan^2 \theta}.$$

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10. (a) Use de Moivre's theorem to show that

[4]

$$\cos 5\theta \equiv 16\cos^5 \theta - 20\cos^3 \theta + 5\cos \theta.$$

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(b) Hence find the five distinct solutions of the equation

[4]

$$32x^5 - 40x^3 + 10x - 1 = 0,$$

giving your answers to 3 decimal places where necessary.

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(c) Hence find the exact value of $\cos \frac{\pi}{15} \cos \frac{7\pi}{15} \cos \frac{11\pi}{15} \cos \frac{13\pi}{15}$.

[2]

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(Total for Question 10 is 10 marks)



Section C n th roots of a complex number

22 marks

11. Express $-2 + 2\sqrt{3}i$ in the form $re^{i\theta}$. [2]

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12. Solve the equation $z^3 = 27i$, giving your answers in the form $re^{i\theta}$. [3]

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13. Solve the equation $z^4 = -16$, giving your answers in the form $a + bi$. [3]

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14. Solve the equation $z^5 = 16 + 16\sqrt{3}i$, giving your answers in the form $re^{i\theta}$. [4]

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15. (a) Express $4\sqrt{2} - 4\sqrt{2}i$ in the form $re^{i\theta}$. [2]

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- (b) Hence solve the equation $z^3 = 4\sqrt{2} - 4\sqrt{2}i$, giving your answers in the form $re^{i\theta}$. [4]

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- (c) Show the points representing your answers to part (b) on an Argand diagram. [2]

- (d) These points are the vertices of a triangle. Find the exact area of the triangle. [2]

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(Total for Question 15 is 10 marks)



Section D Roots of unity and their geometry

19 marks

16. Find the three cube roots of unity, giving your answers in the form $a + bi$. [2]

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17. Given that $\omega = e^{\frac{2\pi i}{5}}$, show that $1 + \omega + \omega^2 + \omega^3 + \omega^4 = 0$. [2]

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18. Solve the equation $z^6 = 1$, giving your answers in the form $e^{i\theta}$. Describe the shape formed by the points representing the roots on an Argand diagram. [3]

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19. The complex number ω is a cube root of unity, with $\omega \neq 1$. [4]

(a) Find the value of $(1 + \omega)(1 + \omega^2)$.

(b) Find the value of $(2 + 5\omega + 2\omega^2)^6$.

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20. The points A , B , C , D , E and F are the vertices of a regular hexagon, labelled anticlockwise, with centre at the origin O of an Argand diagram. The point A represents the complex number $\sqrt{3} + i$.

(a) The vertices of the hexagon are the roots of an equation of the form $z^6 = c$. Find the value of c . [2]

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(b) Find the complex numbers represented by B , C , D , E and F , giving your answers in the form $a + bi$. [3]

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(c) Find the exact area of the hexagon. [2]

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(d) Write down the sum of the six complex numbers represented by the vertices, giving a reason for your answer. [1]

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(Total for Question 20 is 8 marks)



ROAR-LEVEL CHALLENGE

5 marks

21. By considering $\left(z - \frac{1}{z}\right)^5$, where $z = \cos \theta + i \sin \theta$, express $\sin^5 \theta$ in the form $a \sin 5\theta + b \sin 3\theta + c \sin \theta$, where a , b and c are constants to be found. [5]

Hence find the exact value of $\int_0^{\frac{\pi}{2}} \sin^5 \theta \, d\theta$.

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END OF WORKSHEET

Total: 87 marks